THE SECOND REFERENCE

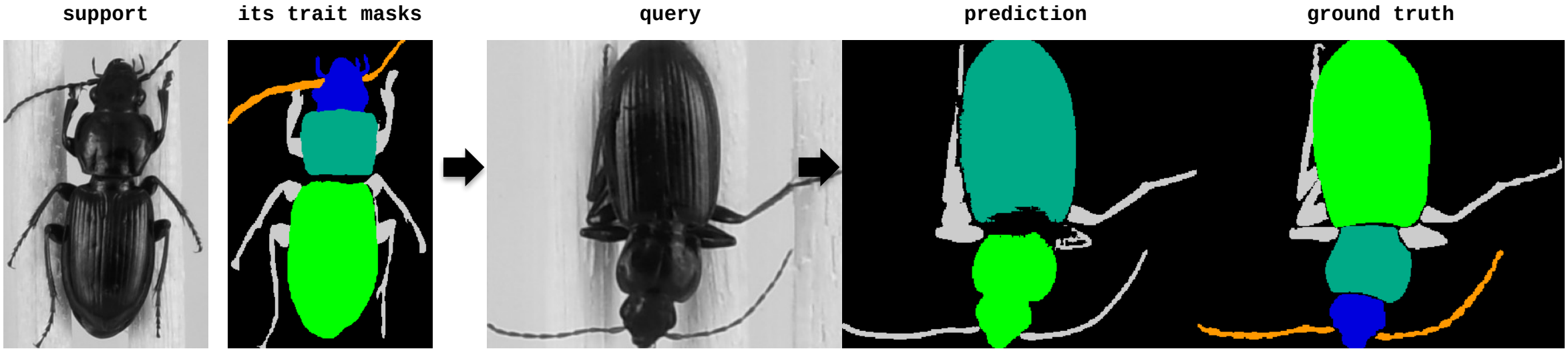
Rotate the support, or label another specimen

implicit: rotate the one you have	**explicit: label a second specimen**	**the answer flips between datasets**

S16 / base SAM 2 / 4 runs / 20 seeds / Hunter Bridges

THE TASK

One labelled specimen, then a query



One labelled mask is the whole supervision budget. Everything in this talk is about spending a second one, or not.

THE QUESTION

Two ways to spend a second reference

IMPLICIT (orientation TTA)





the same specimen, turned 180 degrees, kept outside the memory bank and picked per episode by a confidence rule

cost: 0 labels, 2 forward passes

EXPLICIT (a real 2-shot)





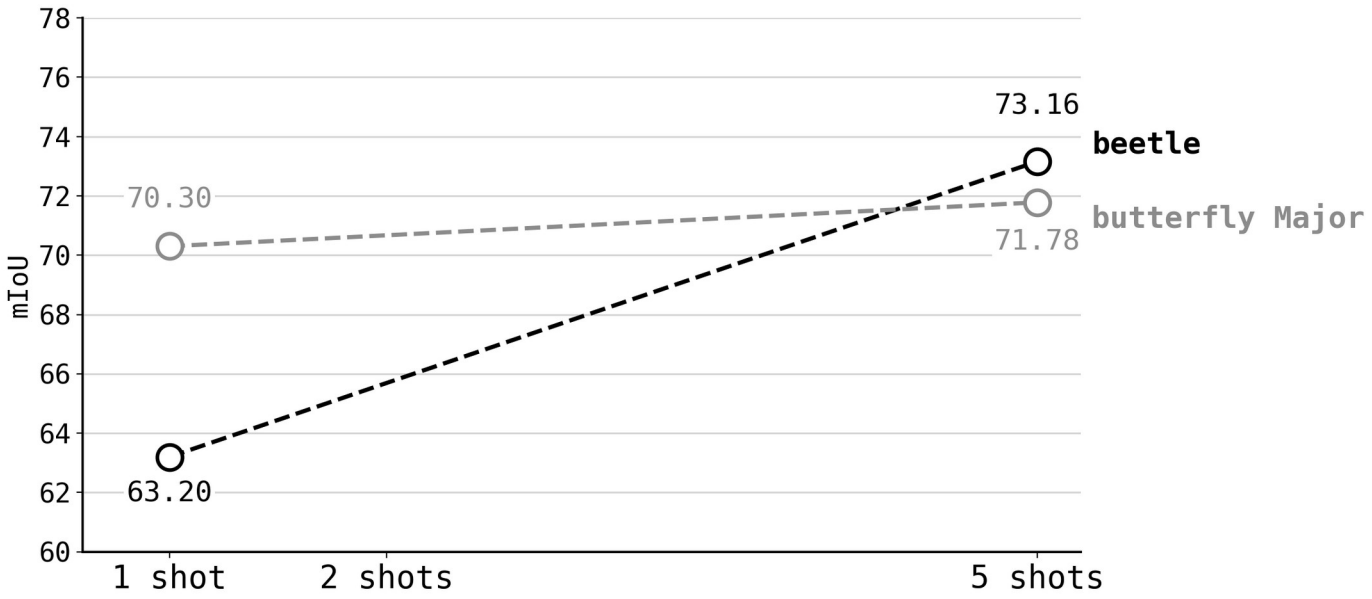
a second individual from the same pool, inside the bank as one 3-frame video, selected by memory attention

cost: 1 label, 1 forward pass

Panels are the project's own S3 render. Since S5c every headline arm has used the implicit version. The explicit one had never been measured at any budget, and it is the cheaper of the two at inference.

WHAT WAS ALREADY MEASURED

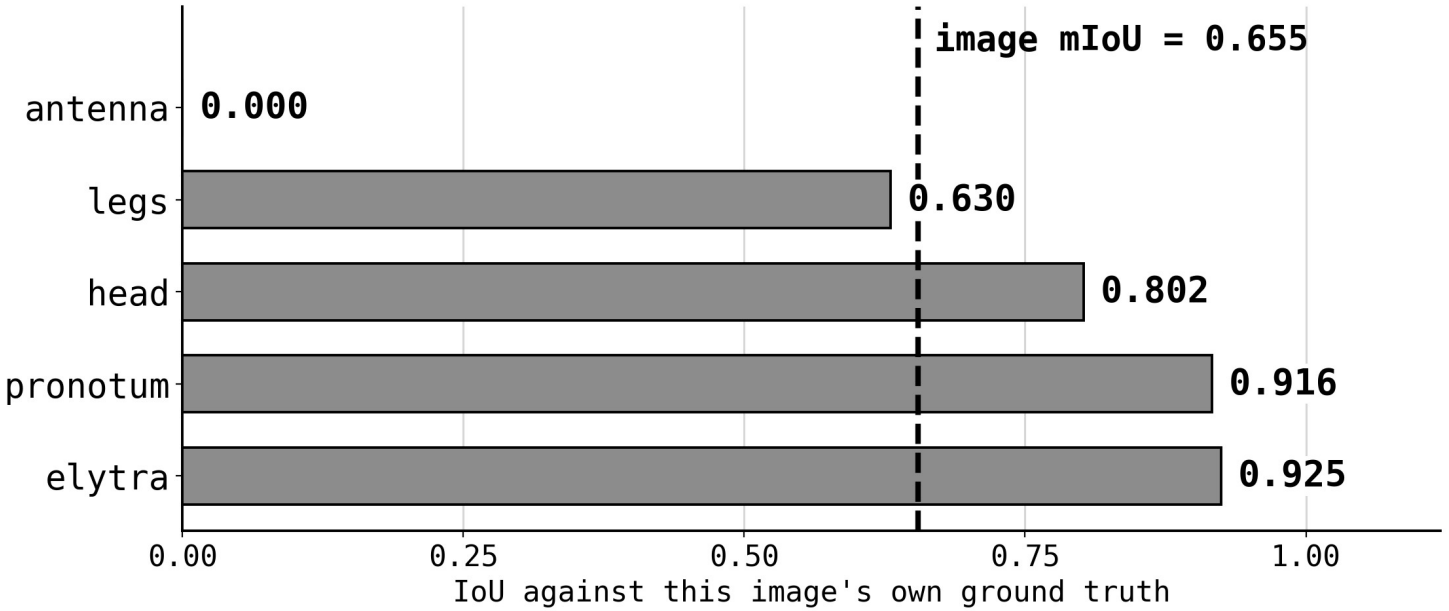
The shot axis had a hole at two



The project has 1-shot and 5-shot on both datasets, never 2. The dashed line is an interpolation nobody measured.

THE GRADING RULE

What one mIoU number means here



Per image: resize logits to the query, argmax over trait channels, IoU per ground-truth trait, then the mean over traits.

A trait the model never claims scores 0 and still counts, so a missed antenna costs as much as a botched elytra.

DESIGN

Four runs, two of them anchors

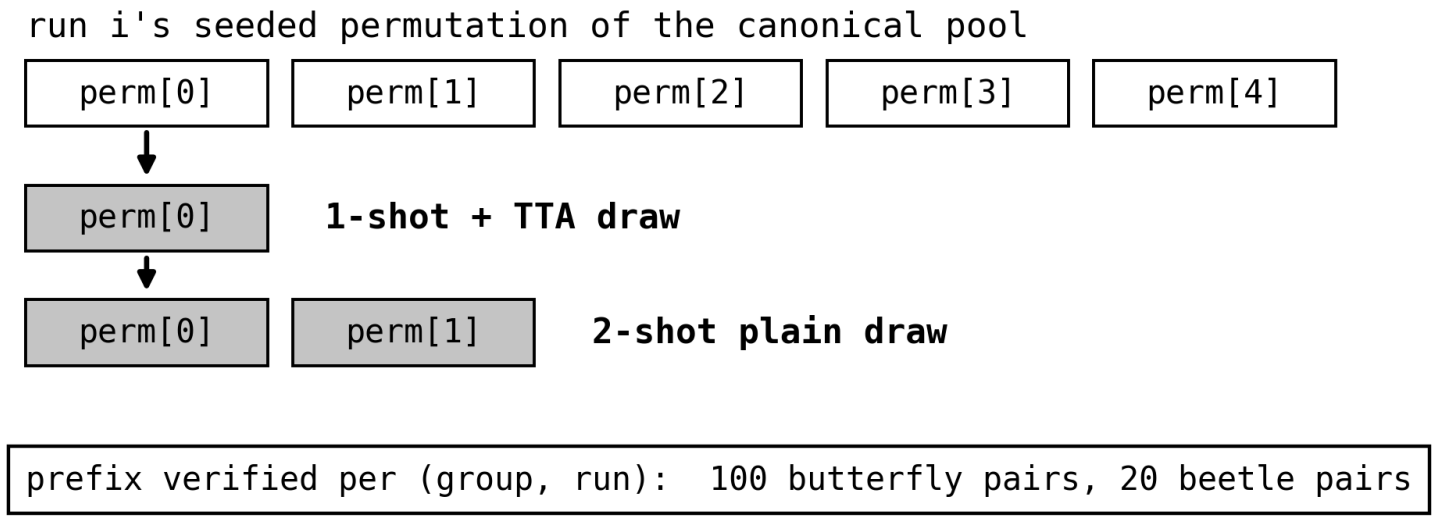
	1 shot + TTA	2 shots, plain
butterfly Major	**75.18** resumed from S11, pinned	**71.48** new measurement
beetle	**63.86** resumed from S11, pinned	**69.04** new measurement

base SAM 2, no adapters, 20 seeds each, canonical protocol per dataset

The two TTA cells resume the banked S11 cells, and the gate pins them to 75.18 and 63.86 within 0.5.

That certifies the wiring of the merge, not TTA itself. The two 2-shot cells are the only new measurement here.

Why this comparison is allowed

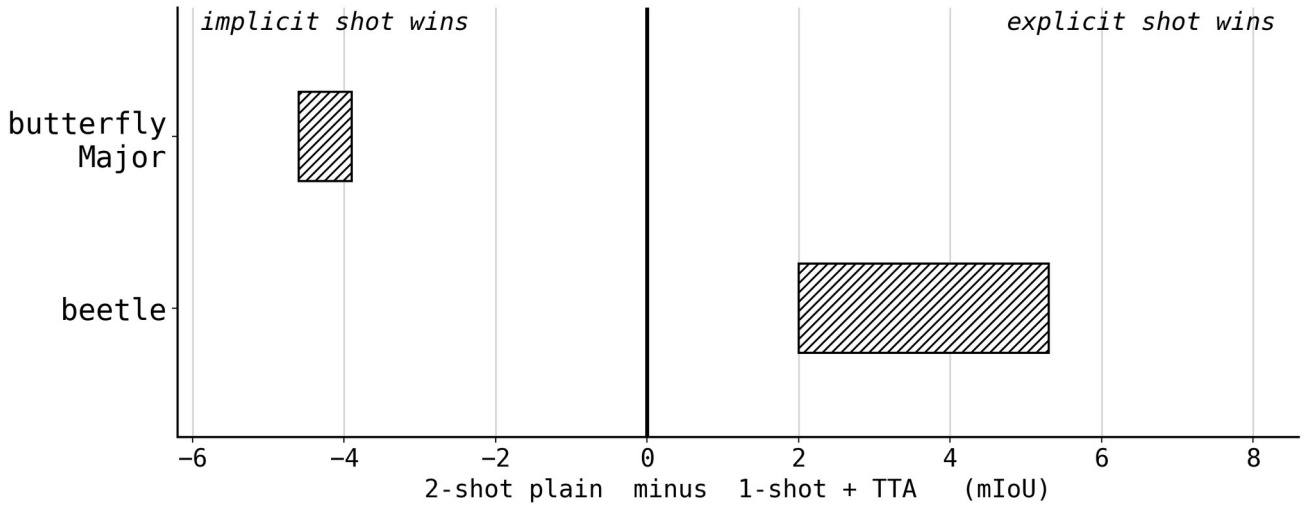


The row crosses protocol.shots and protocol.support_set_hash, both comparability fields, so registry.compare refuses it.

The licence is a property of the draws: both cells slice a prefix of one seeded permutation, so the 1-shot support is the first element of the 2-shot pair and the shared support's difficulty still cancels.

WRITTEN BEFORE THE RUN

Two predictions, opposite signs

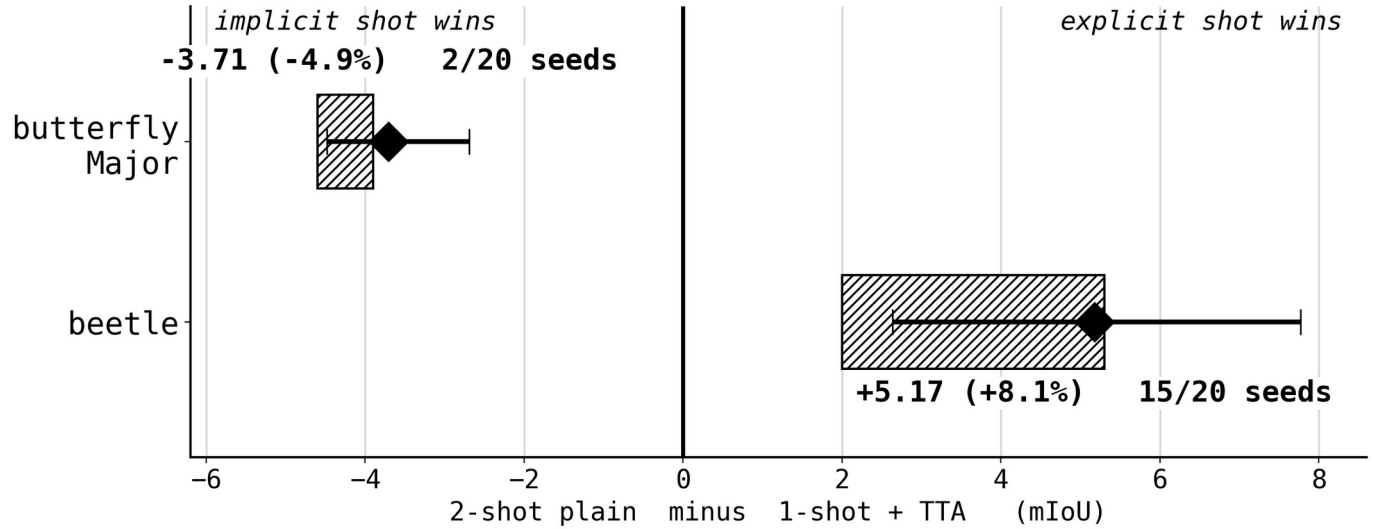


Butterfly, band -3.9 to -4.6: Major's defect is query orientation, and 37% of cyrbia's test specimens are pinned upside down, so no upright second specimen reaches it.

Beetle, band +2 to +5.3: the 5-shot mechanism is species diversity, and a rotated copy of one specimen adds no species.

RESULT

The mirror image lands, both signs



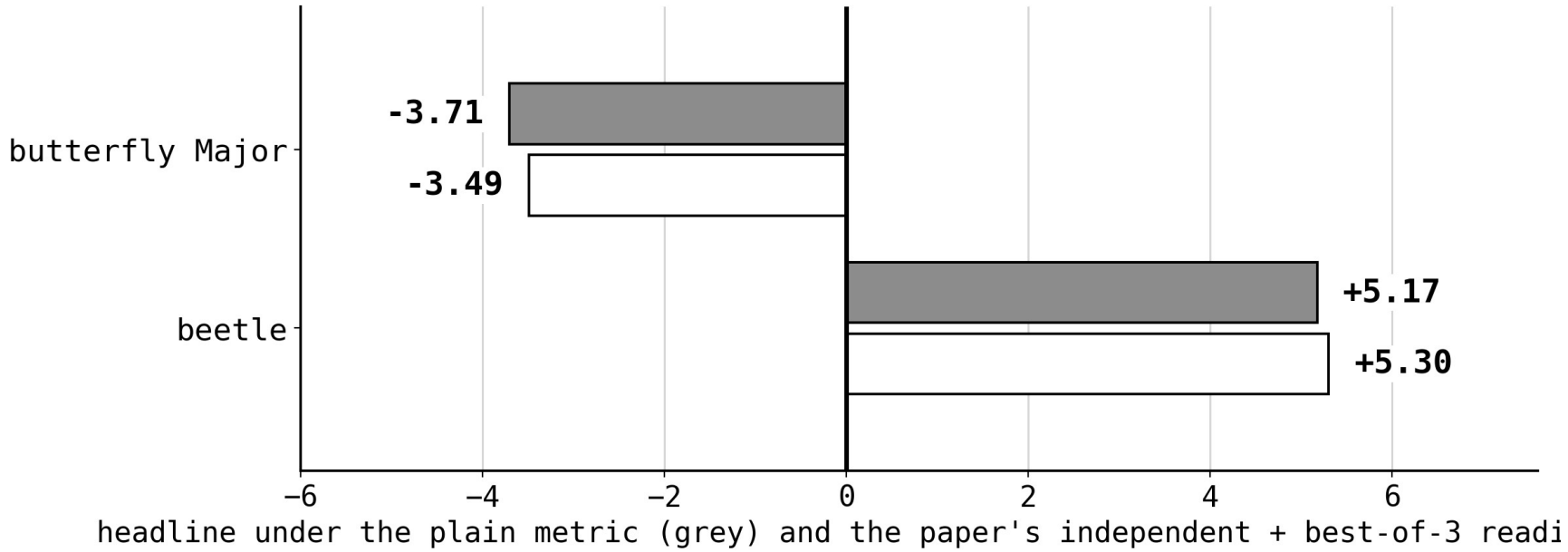
butterfly 75.18 vs 71.48, -3.71 (-4.9%), significant, 0.19 short of the band edge

beetle 63.86 vs 69.04, +5.17 (+8.1%), significant, at the top of its band

Gate PASS 14 of 14. The anchors resume S11's cells; the 2-shot cells are new.

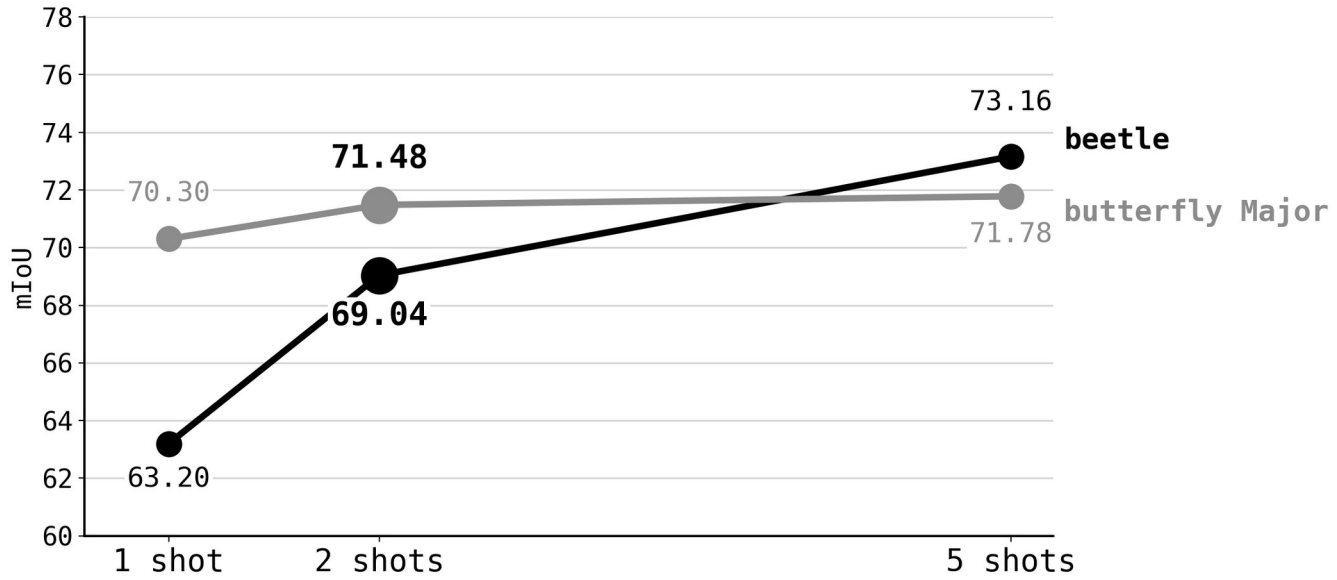
ROBUSTNESS

The sign survives the other reading



Under the paper's full reading, independent overlap plus boundary best-of-3, the headlines are -3.49 and +5.30, both still significant. That reading changes two things at once, and neither sign moves.

Major's shot curve is flat by two

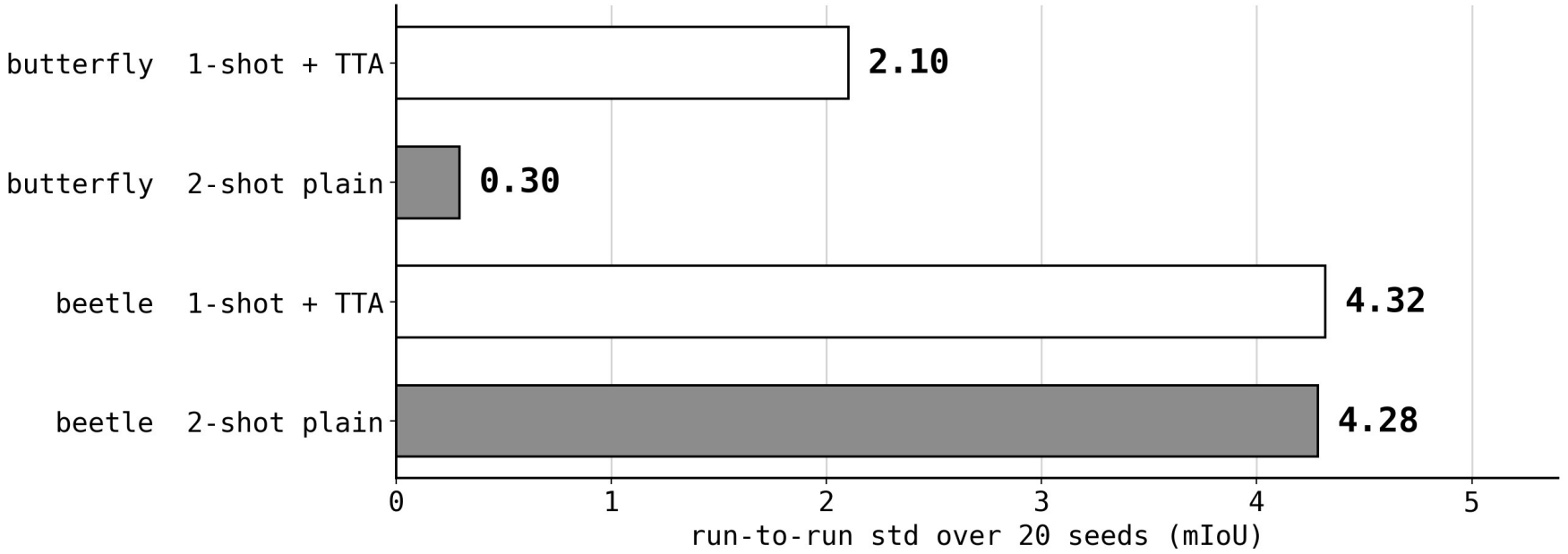


2-shot came in at 71.48 against a predicted 70.6 to 71.3, so the band miss is on the 2-shot side.

+1.18 of the measured +1.48 five-shot lift, 80% of it, arrives with the first added specimen. Unpaired against the banked level.

BUTTERFLY MAJOR

The second specimen bought spread

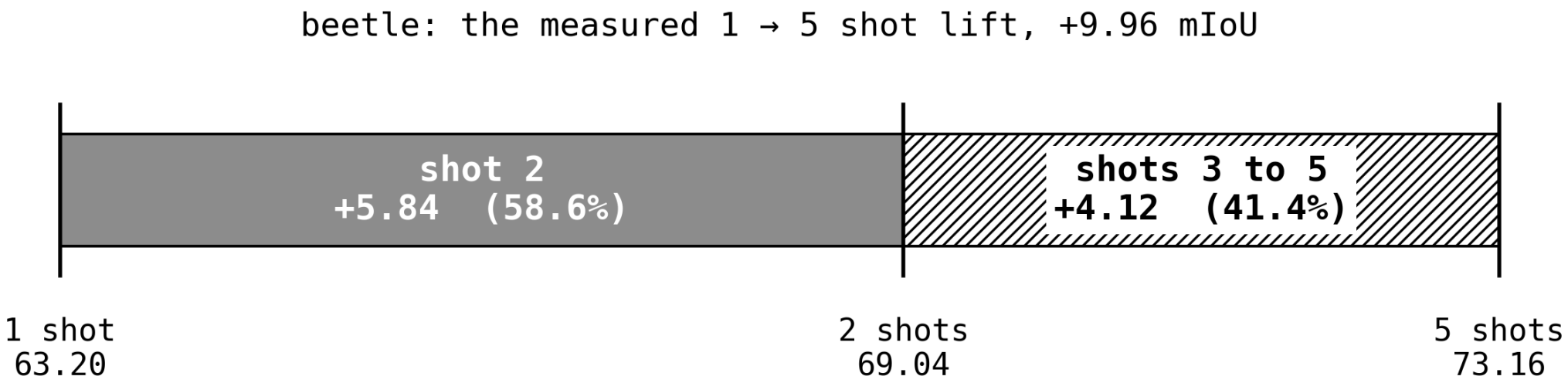


std 2.10 to 0.30, a drop of 86%, the tightest cell measured at 1 or 2 shots. The level stayed 3.71 below TTA.

That is what S5b and S6 said support-side levers do on Major. The residual is query-side, and support-side levers do not reach it.

BEETLE

Beetle earns the lift on shot two

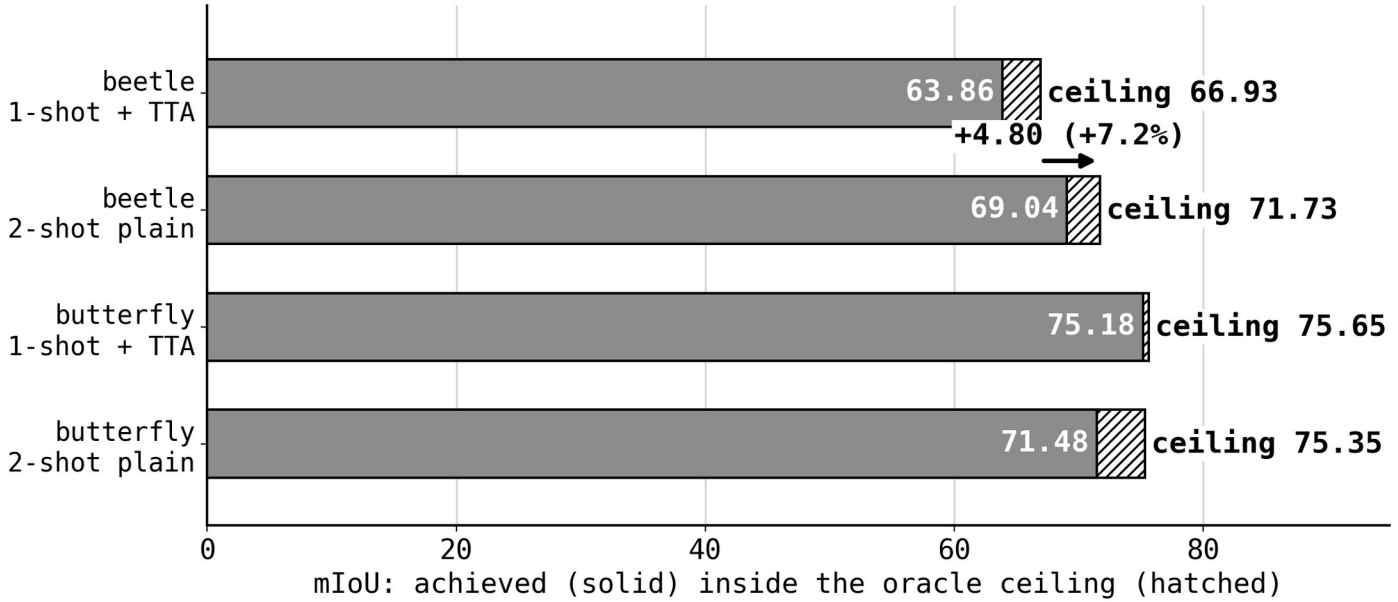


+5.84 (+9.2%) over the banked 63.20, unpaired, which is 58.6% of the whole 1 to 5 shot lift from one added specimen.

Concave as predicted, but past the top of the pre-registered 35 to 55% share band. Beetle overshot where butterfly undershot.

BEETLE

It changed what gets proposed

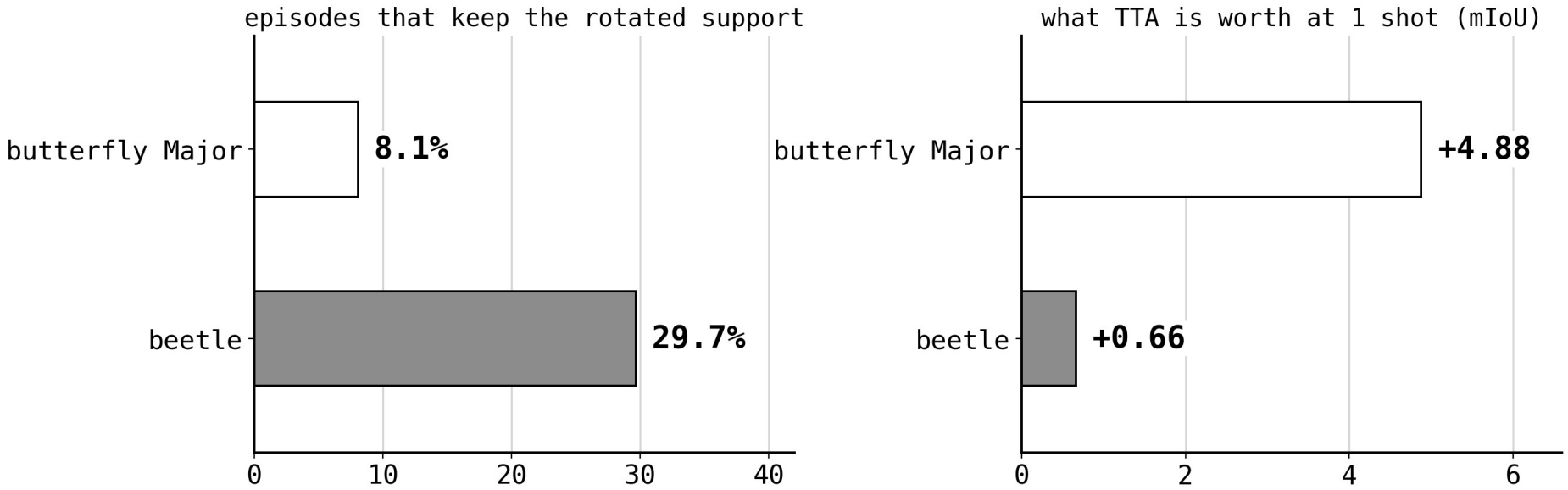


beetle oracle ceiling 66.93 to 71.73, +4.80 (+7.2%). The second specimen changed the proposals, not just the assignment.

Butterfly's ceiling is flat within a third of a point, 75.65 to 75.35, with no paired test on the oracle reading. S14sam3's arch arm moved this axis by training; here it moved by data.

MECHANISM

TTA is an orientation shot only

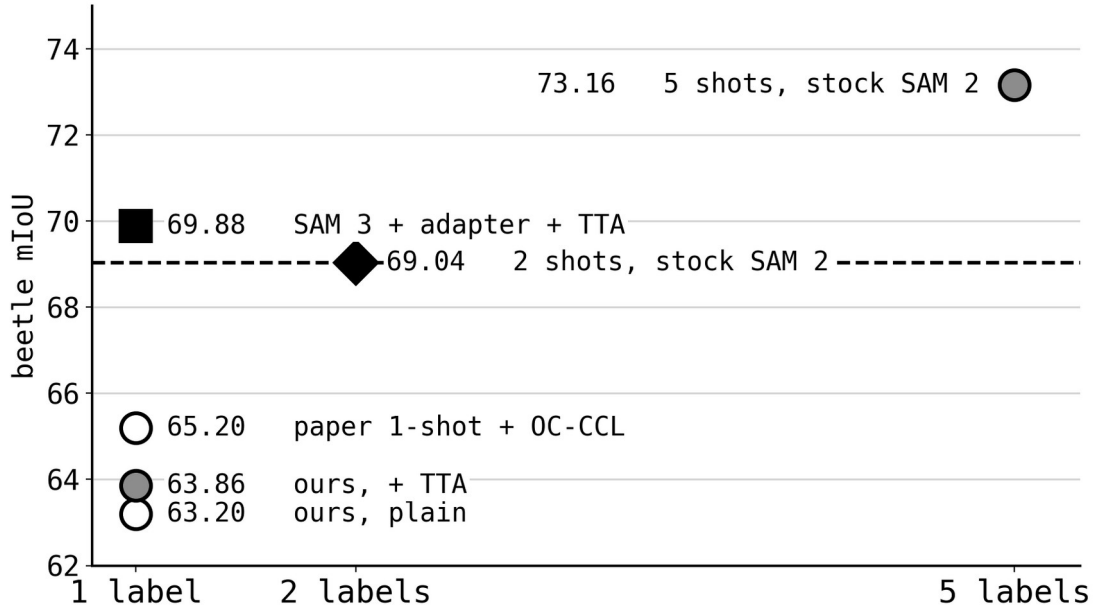


On beetle the rotated support is kept in 29.7% of episodes and buys +0.66 (+1.0%). The explicit specimen is worth about 9 times that, because a rotated copy adds no species.

On Major it fires in 8.1% of episodes and is worth +4.88 (+6.9%), because those are the rotated queries nothing else reaches.

WHAT IT COSTS

What the second label actually buys



One extra labelled mask on stock SAM 2 puts beetle at 69.04, within noise of the project's champion at one label, 69.88, which is a model swap plus a trained adapter plus TTA.

On SST's own label axis the cheapest training-free point above the paper's 1-shot OC-CCL row of 65.2 is now 2 labels.

WHAT IT SUGGESTS

Three things this changes

01 The two datasets are mirror images, each by the mechanism its own residual named. Neither result generalises to the other.

02 Ask what a technique adds, not how many references it looks like. TTA adds an orientation, not a specimen.

03 A second label is the strongest per-label lever measured on beetle, and it needs no training at all. 5-shot is still larger.

S16, gate PASS 14 of 14, landed 2026-09-01. Every number here is in results/s16.zip.

BACKUP

Every cell, as recorded

	1-shot + TTA	2-shot plain	headline
butterfly plain	75.18 +/- 2.10	71.48 +/- 0.30	-3.71 (-4.9%)
butterfly best-of-3	76.72	73.23	-3.49
butterfly oracle	75.65	75.35	-0.30
beetle plain	63.86 +/- 4.32	69.04 +/- 4.28	+5.17 (+8.1%)
beetle best-of-3	65.45	70.75	+5.30
beetle oracle	66.93	71.73	+4.80

20 seeds per cell. CIs on the plain rows: butterfly [-4.48, -2.69], 2 of 20 seeds improved; beetle [+2.63, +7.77], 15 of 20. The oracle rows are differences, with no paired test run on that reading.

BACKUP

What the gate checks, per dataset

01	both cells on the dataset's canonical protocol, 20 of 20 seeds each
02	the 1-shot TTA anchor reproduces the S11 ledger within 0.5
03	the headline carries its per-run prefix-nesting proof
04	the headline declares the model.tta toggle it crosses
05	2-shot clears the banked 1-shot level, so no conditioning frame is dropped
06	the paper's best-of-3 ran alongside the plain metric
07	the headline itself is recorded with its CI and seed count

Seven per dataset, 14 of 14 PASS. Naming a protocol field in ignore_fields is still refused; the relaxation must arrive with its evidence.